MEASURING COMPLEXITY OF DYNAMICAL COMPLEX SYSTEMS

To-do

- In depth study of each measure, see `ref` folder for paper PDF. Add more measures if needed.
- Study how each measure can be applied to each dynamical complex system.
  - Applicability – e.g. TL can be applied to rule level, not pattern level?
  - Discrete measures – need to discretized / coarse-grained some continuous systems (particles, cont CA) for using discrete measures?
  - Implementation details – e.g. use LZ or PNG compression as proxy of AC and LD.
- [Bert] complete and share *Lenium* (Lenia Python library) for easier coding. Prepare markdown for coding agent.

1 Complexity Measures

bold = recommended

1. **AC = Algorithmic Complexity (Kolmogorov-Chaitin) / Algorithmic Information Theory**
2. **SE = Shannon Entropy / Information-Theoretic Complexity**
3. **LE = Lyapunov Exponents**
4. FD = Fractal Dimension (Hausdorff, Mandelbrot)
5. EC = Effective Complexity (Gell-Mann)
6. SC = Statistical Complexity (Crutchfield)
7. **LD = Logical Depth (Bennett)**
8. TD = Thermodynamic Depth (Lloyd, Pagels)
9. **TL = Transient Length (Hudcová, Mikolov)**
10. **PID = Partial Information Decomposition (Williams, Beer)**
11. FID = Functional Information Decomposition (Bohm)
12. **TE = Transfer Entropy (Schreiber)**
13. IIT = Integrated Information Theory (Tononi)
14. **ITI = Information Theory of Individuality (Krakauer)**
15. AT = Assembly Theory (Cronin, Walker)
16. FI = Functional Information (Szostak, Hazen)
17. LL = Langton's Lambda Parameter
18. WZ = Wuensche's Z Parameter

1.1 Granularity level

- pattern level (self-organization) = **ITI, LD, AC**, FD, EC, TD, AT
- rule level (edge of chaos) = **LE, TE, TL**, SC, FI, LL, WZ
- system level (expressiveness) = **SE, PID**, FID, IIT

1.2 Compatibility with dynamical complex systems

local = capture local interactions; dynamical = capture temporal dynamics

- local and dynamical = **ITI, TE, PID**, FID, IIT

- global and dynamical = **LE, TL, LD**, TD, SC, FI
- global and static = **AC, SE**, FD, EC, AT, LL, WZ

1.3 Mentions

- Mitchell 2009 – Complexity: A Guided Tour (p. 94)
  - AC, SE, EC, LD, TD, SC, FD

2 Details

Measure Randomness and Chaos

- equate complexity with unpredictability, incompressibility, or geometrical scattering

2.1 AC = Algorithmic Complexity (Kolmogorov-Chaitin) / Algorithmic Information Theory

- reference:
  - Kolmogorov 1963 https://doi.org/10.1016/S0304-3975(98)00075-9
  - Chaitin 1977 https://doi.org/10.1147/rd.214.0350
  - Zenil 2020 https://doi.org/10.3390/e22060612
  - Zenil et al. 2022 [book] https://doi.org/10.1007/978-3-662-64985-5
- type: pattern, global, static
- quality: measures compressibility
- quantity: minimum program length to generate pattern

$$K(x) = \min\{|p| : U(p) = x\}$$

  where $|p|$ is the length of the shortest program p on a universal Turing machine U that outputs string x.
- non-computable, approximated by compressed size
- peak: pure randomness, data cannot be further compressed
- related: **AP = Algorithmic Probability (Solomonoff-Levin)**
- related: **LZC = Lempel-Ziv Complexity**
  - Lempel and Ziv 1976 https://doi.org/10.1109/TIT.1976.1055501
  - for Lenia: Adachi et al. 2024 https://arxiv.org/abs/2408.06374

2.2 SE = Shannon Entropy / Information-Theoretic Complexity

- reference:
  - Shannon 1948 https://doi.org/10.1002/j.1538-7305.1948.tb01338.x
  - python: scipy.stats.entropy https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.entropy.html
- type: system / pattern, global, static
- quality: measures unpredictability
- quantity: average amount of information produced by a stochastic source of data

$$H(X) = -\sum_i p_i \log_2 p_i$$

  where p_i is the probability of the system being in state i.
- peak: pure randomness, every outcome is equally likely and highly uncertain

2.3 LE = Lyapunov Exponents

- reference:
  - Lyapunov 1992 https://doi.org/10.1080/00207179208934253
  - python: import lyapynov https://github.com/ThomasSavary08/Lyapynov
- type: rule, global, dynamic

- quality: measures chaos
- quantity: rate of nearby trajectories diverge over time

$$\lambda = \lim_{t\to\infty} \frac{1}{t} \ln \left| \frac{\Delta x(t)}{\Delta x(0)} \right|$$

  where $\Delta x(t)$ is the separation between two trajectories at time t relative to their initial small separation $\Delta x(0)$.
- peak: maximum chaos, perfectly unpredictable dynamic system
- related: **KYD = Kaplan-Yorke Dimension**
  - for Lenia: Yevenko et al. 2026 https://doi.org/10.1162/ISAL.a.834
  - python: https://github.com/iyevenko/DynamicalCA

2.4 FD = Fractal Dimension (Hausdorff, Mandelbrot)

- reference:
  - Mandelbrot 1967 https://doi.org/10.1126/science.156.3775.636
  - python: import fractal https://github.com/wanglab-georgetown/fractal
- type: pattern, global, static
- quality: measures geometric complexity
- quantity: how much detail a shape retains when zoom in; geometric roughness or space-filling capacity

$$D = \lim_{\epsilon\to 0} \frac{\log N(\epsilon)}{\log(1/\epsilon)}$$

  where $N(\epsilon)$ is the number of boxes of side length ϵ required to cover the geometric pattern.
- peak: pure randomness, fills maximum space (white noise > fractal)

Measure Structural Complexity

- isolate meaningful structure from raw noise

2.5 EC = Effective Complexity (Gell-Mann)

- references:
  - Gell-Mann and Lloyd 1996
  - https://doi.org/10.1002/(SICI)1099-0526(199609/10)2:1%3C44::AID-CPLX10%3E3.0.CO;2-X
- type: pattern, global, static
- quality: measures regularities
- quantity: ignore random and noisy features, calculate algorithmic complexity of meaningful signal

$$\mathcal{E}(x) = K(R_x)$$

  where $K(R_x)$ is the algorithmic complexity of the regularities (the extracted "schema" or rule R) of dataset x, ignoring the random noise.
- non-computable
- peak: edge of chaos, high regularity + non-trivial structure

2.6 SC = Statistical Complexity (Crutchfield)

- references:
  - Crutchfield and Young 1989 https://doi.org/10.1103/PhysRevLett.63.105
  - Crutchfield and Feldman 1996 https://doi.org/10.48550/arXiv.cond-mat/9702191
  - python: import emic https://github.com/johnazariah/emic
- type: rule, global, dynamic
- quality: measures memory
- quantity: minimum memory needed for epsilon-machine to predict statistical future of system

$$C_\mu = -\sum_S P(S) \log_2 P(S)$$

  where $P(S)$ is the probability of the system being in causal state S within its minimal epsilon-machine (hidden Markov model).
- peak: edge of chaos, system governed by intricate hidden states

2.7 LD = Logical Depth (Bennett)

- references:
  - Bennett 1988 https://doi.org/10.1093/oso/9780198537748.003.0008
- type: pattern, global, dynamic
- quality: measures computation history
- quantity: execution time of shortest program, low at order (regular) and random (straight print)

$$D(x) = \min\{\text{time}(p) : U(p) = x, |p| \approx K(x)\}$$

  where time(p) is the execution time of the shortest possible program p that generates pattern x.
- non-computable, approximated by decompression execution time
- peak: edge of chaos, require massive computation to unfold

2.8 TD = Thermodynamic Depth (Lloyd, Pagels)

- references:
  - Lloyd and Pagels 1988 https://doi.org/10.1016/0003-4916(88)90094-2
- type: pattern, global, dynamic
- quality: measures causal history
- quantity: entropy produced during the physical history of a state's formation

$$D_r = \bar{S} - S_0$$

  where $\bar{S}$ is the coarse-grained entropy corresponding to macroscopic state, and S_0 is the fine-grained entropy corresponding to microscopic state.
- peak: edge of chaos, system with long and complex causal history

2.9 TL = Transient Length (Hudcová, Mikolov)

- references:
  - for ECA: Hudcová and Mikolov 2020 https://doi.org/10.1162/isal_a_00260
- type: rule, global, dynamic
- quality: measures transient time
- quantity: how many simulation steps a dynamical system takes to fall into a stable attractor

$$t_u = \min\{t : F^t(u) \approx F^{t+T}(u)\}$$

  where $F^t(u)$ is the global update rule F applied t times to the initial configuration u, and T is the period of the final stable loop or fixed attractor the system eventually falls into.
- peak: edge of chaos, very long transient periods before settling

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Measure Causal Flow and Synergy

- decompose how components communicate and influence one another locally, map complex wiring

2.10 PID = Partial Information Decomposition (Williams, Beer)

- references:
  - Williams and Beer 2011 https://doi.org/10.48550/arXiv.1004.2515
  - python: import dit https://dit.readthedocs.io/en/latest/measures/pid.html
- type: system, local, dynamic
- quality: measures multivariate information interactions
- quantity: breakdown of mutual information into redundant, unique, and synergistic components

$$I(Z; X, Y) = U(Z; X) + U(Z; Y) + R(Z; X, Y) + S(Z; X, Y)$$

  where the total mutual information I is decomposed into unique (U), redundant (R), and synergistic (S) information provided by sources X and Y about target Z.
- peak: synergistic networks, components interact to create new information

2.11 FID = Functional Information Decomposition (Bohm)

- references:
  - Bohm et al. 2025 https://arxiv.org/abs/2509.18522
  - python: import fid-tools pypi.org/project/fid-tools/
- type: system, local, dynamic
- quality: measures functional relationships within complete systems
- quantity: distribution of synergistic vs redundant mappings in complete input-output function

$$I_{\text{tot}}(Y; X) = I_{\text{syn}}(X \to Y) + \sum_i I_{\text{ind}}(Y; X_i)$$

  where the total information I_{tot} is decomposed into synergistic information I_{syn} and individual information I_{ind} provided by each individual input variable X_i alone.
- peak: complex task execution, networks that optimally distribute computational roles

2.12 TE = Transfer Entropy (Schreiber)

- references:
  - Schreiber 2000 https://arxiv.org/abs/nlin/0001042
  - python: import infomeasure https://github.com/cbueth/infomeasure
- type: rule / pattern, local, dynamic
- quality: measures information flow
- quantity: directed, asymmetric exchange of information between two coupled components

$$T_{X \to Y} = H\left(Y_t \mid Y_{t-1:t-L}\right) - H\left(Y_t \mid Y_{t-1:t-L}, X_{t-1:t-L}\right)$$

  where X and Y are coupled variables and H is Shannon entropy, measuring how much knowing the past of X ($X_{t-1:t-L}$) reduces uncertainty about the future of Y (Y_t) given its own past ($Y_{t-1:t-L}$).
- peak: edge of chaos, components exchange structured, meaningful signals

2.13 IIT = Integrated Information Theory (Tononi)

- references:
  - Tononi 2004 https://doi.org/10.1186/1471-2202-5-42
  - Oizumi et al. 2014 https://doi.org/10.1371/journal.pcbi.1003588
  - Albantakis et al. 2023 https://doi.org/10.1371/journal.pcbi.1011465
- type: system, local, dynamic
- quality: measures consciousness and integrated complexity
- quantity: causal information generated by whole system above its independent parts

$$\Phi = D(p(AB||\text{int}) \parallel p(A \times B||\text{part}))$$

  where big phi Φ is the distance between $p(AB||\text{int})$ the intact cause-effect repertoire or probability distribution of the whole system AB, and $p(A \times B||\text{part})$ the partitioned distribution where information flow between subsystem A and subsystem B is severed.
- peak: consciousness and brains, optimal functional integration; low in random networks or rigid grids

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Measure Agency, Selection, Function

- quantify how macroscopic structures persist as distinct individuals and resist environmental decay

2.14 ITI = Information Theory of Individuality (Krakauer)

- references:
  - Krakauer et al. 2020 https://doi.org/10.1007/s12064-020-00313-7
  - for mutual information – python: import infomeasure https://github.com/cbueth/infomeasure
- type: pattern, local, dynamic

- quality: measures biological individuality and persistence
- quantity: how much an entity propagates its own information forward in time and predicts its own future state vs environment dictation
  - Organismal Individuality (Endogenous Determination):
    $$A^* = I(S_{n+1}; S_n)$$
  - Colonial Individuality:
    $$A = I(S_{n+1}; S_n \mid E_n)$$
  - Environmental Determined Individuality (Driven Form):
    $$n_C = I(S_{n+1}; E_n \mid S_n)$$
    where the system S_n and the environment E_n at time step n.
- peak: autonomous agents, e.g. living organisms, bounded individuals

2.15 AT = Assembly Theory (Cronin, Walker)

- references:
  - Sharma et al. 2023 https://doi.org/10.1038/s41586-023-06600-9
  - Kempes et al. 2025 https://doi.org/10.1038/s44260-025-00049-9
  - python: import assembly-theory https://github.com/DaymudeLab/assembly-theory
  - python: https://doi.org/10.5281/zenodo.8017327
- type: pattern, global, static
- quality: measures physical complexity and evidence of selection
- quantity: assembly index = minimum historical steps required to construct an abundant object
  $$A = \sum_{i=1}^{N} e^{a_i} \left(\frac{n_i - 1}{N_T} \right)$$
  where the total assembly A is the sum of exponent of assembly index a_i weighted by its copy number n_i.
- peak: selection and life, require evolutionary pressure

2.16 FI = Functional Information (Szostak, Hazen)

- references:
  - Hazen et al. 2007 https://doi.org/10.1073/pnas.0701744104
  - Wong et al. 2023 https://doi.org/10.1073/pnas.2310223120
- type: rule, global, dynamic
- quality: measures functional complexity
- quantity: rarity of specific working configuration compared to all possible random arrangements
  $$I(E_x) = -\log_2 \left(\frac{M(E_x)}{N} \right)$$
  where $M(E_x)$ is the number of specific configurations that meet or exceed the functional threshold E_x, out of all possible random configurations N.
- peak: functional optimization, e.g. highly specific enzymes or tools

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Measure Rule-Space Phase Topologies

- evaluate discrete lookup tables to predict emergence of complex behaviors, not dynamic patterns

2.17 LL = Langton's Lambda Parameter

- references:
  - Langton 1990 https://doi.org/10.1016/0167-2789(90)90064-V
  - python: import cellpylib https://github.com/lantunes/cellpylib

- type: rule, global, static
- quality: measures living rules
- quantity: fraction of rule transitions in lookup table that lead to a non-quiescent state

$$\lambda = \frac{K^N - n_q}{K^N}$$

  where K is the number of possible states, N is the number of cells in a neighborhood, n_q is the number of transition rules that map to the quiescent state (q).
- peak: edge of chaos, at critical threshold lambda_c

2.18 WZ = Wuensche's Z Parameter

- references:
  - Wuensche 1994
  - https://doi.org/10.1002/(SICI)1099-0526(199901/02)4:3<47::AID-CPLX9>3.0.CO;2-V
- type: rule, global, static
- quality: measures phase space convergence
- quantity: probability of a future state uniquely determines its pre-image
- peak: edge of chaos, balance of freezing vs scattering

3 Dynamical Complex Systems

= many components + local interactions + evolution over time

- Cellular automata
  - Elementary CA – 2 neighbors
  - von Neumann CA – 4 neighbors
  - Moore CA – 8 neighbors, including Conway's Game of Life
  - Continuous CA – Asymptotic Lenia, with low or medium resolution
- Particle swarms
  - Boids
  - Swarm Chemistry
  - Particle Life
  - Particle Lenia